

Regularized Multivariate Two-way Functional Principal Component Analysis

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Outline

- 1 Introduction & Background
- 2 A SVD Approach for Regularized Multivariate FPCA
- 3 Two-way Regularized Multivariate FPCA
- 4 Conclusion & Future Work

Introduction

Functional Data Analysis:

Functional data are observations that vary continuously over a domain (like time, space, or wavelength) and are often visualized as curves, trajectories, or functions rather than isolated points.

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Functional Data Analysis (FDA) is a statistical framework that treats these observations as realizations of smooth underlying functions, allowing for more accurate modeling and interpretation of continuous processes.

Regularized MFPCA

Two-way Regularized MFPCA

Conclusion & Future Work